Spectrum Sensing System Design

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Selected SDR: This document focuses on a network-attached SoC option (AD9361 + Zynq-7020 family) and the supporting system design. For comparison only, note that host-attached USB 3.0 SDRs and other form factors exist; they are intentionally not included in the bill-of-materials (BOM) or core hardware sections of this report to avoid vendor-specific BOM entries. The design below therefore centers on the network-attached 7020-class device and generic host compute.

7020-SDR (AD9361 + Xilinx Zynq-7020):

- Covers ~70 MHz to 6 GHz (VHF, UHF, C-band).
- Integrated Xilinx Zynq-7020 SoC (Dual ARM + FPGA) for real-time DSP offload.
- Gigabit Ethernet interface (network-attached, libiio compatible) — enables long-distance remote deployment and PoE possibility.
- Fully compatible with `libiio`, `PyADI-IIO`, GNU Radio, and SigMF.
- Price examples: 180 • – 220 •

Recommended compute node: Jetson Orin Nano / Jetson-class device:

- GPU acceleration (e.g., 128 CUDA cores) for parallel tasks (FFT, ML).
- Supports TensorFlow / PyTorch for signal classification and anomaly detection.
- Compact and energy-efficient for edge or rooftop gateway deployments.
- Price example: 269 (275 TOPS). •

Antenna System Configurations:

- *Channel 1:* VHF/UHF Antenna (70 MHz – 3 GHz)
- *Channel 2:* C-Band Antenna (4 GHz – 6 GHz)

Band	**Antenna Type**	**Model Example**	**Frequency Range**	**Gain**
VHF/UHF	Discone	Comet DS-150B	25 MHz – 1.3 GHz	2–3 dBi
VHF/UHF	Log-Periodic	HyperLOG 7060	700 MHz – 6 GHz	5–7 dBi
C-Band	Horn Antenna	Pasternack PE9851	4–8 GHz	10–15 dBi
C-Band	Helical	Custom/DIY	4–6 GHz	12–14 dBi

Table 1: Antenna selection for multi-antenna system

Recommendation: add a wideband log-periodic (e.g., HyperLOG 4080, 400 MHz–8 GHz). Estimated accessory costs: 476 + SMA 5 m cable 106 + SMA-to-N adapter 28. •

Supporting Hardware

i) **Low Noise Amplifiers (LNAs)** Note: SAWbird+ variants include band-specific SAW

Model	Range	Gain	NF
Nooelec SAWbird+ 85 •	Multi-band (variant)	20 dB	<1 dB
Mini-Circuits ZX60-33LN-S+ 159 •	50 MHz – 3 GHz	20 dB	3 dB
SparkGap LNA4ALL	28 MHz – 2.5 GHz	20 dB	1.5 dB

filters; select the correct variant per band or use wideband LNAs with external filters.

ii) **Automatic Antenna Switching System**

- RF switch examples: Mini-Circuits ZSWA-4-30DR+ (DC–3 GHz, 4-port) or Koarmy QM-SW-4S (0.1–6 GHz). Control via Jetson GPIO or a small MCU.
- Preselection (protect front-end / prevent overload): FM broadcast notch (88–108 MHz), bandpass tiles for 700–960 MHz, 1.7–2.7 GHz, 3–4 GHz, 4–6 GHz.
- Stable timing/clocking: GPSDO or external 10 MHz + PPS for frequency accuracy and multi-node coherence (TDOA).
- Digitally-controlled step attenuator (0–31 dB) to avoid ADC saturation in strong-signal environments.

iii) **Outdoor / Rooftop Monitoring**

- Use non-penetrating roof mounts or standard mast mounting (1.25"–2").
- Weatherproof enclosures and grounding / lightning protection.
- Low-loss coax (LMR-400) for longer cable runs to minimize insertion loss.
- For network-attached SDRs, use shielded Cat6 and PoE to simplify rooftop deployments (100 m runs without active USB repeaters).

Practical recommendation:

- LNAs: Nooelec SAWbird+ for narrow-band needs or Mini-Circuits ZX60 for wideband low-NF amplification.
- Interface: prefer network-attached SDRs (GigE / libiio) for remote rooftop installations; host-attached USB 3.0 SDRs are viable for local/short-run deployments but are excluded from BOM entries here.

Testing & Calibration

libiio-utils: Use `iio_info` and `iio_readdev` to validate device reachability and streaming for network-attached SDRs.

VSWR / Antenna: Use a NanoVNA or equivalent to verify antenna matching.

Frequency Sweep: Sweep available frequency ranges to characterize front-end response and identify spurs.

Noise Floor: Measure baseline noise levels for sensitivity and detection threshold tuning.

Software Tools

GNU Radio: Flowgraph development, spectrum estimation, demodulation blocks.

GQRX: Quick spectrum visualization and tuning.

URH (Universal Radio Hacker): Protocol analysis and decoding.

PyADI-IIO / libiio: For Python control and streaming with AD9361-based network SDRs.

Vendor / host drivers: Use appropriate host drivers or vendor-provided libraries for device control and streaming.

Compute and processing pipeline

1. ML pipeline: Use PyTorch/TensorFlow on Jetson (or Orin) for classification; leverage cuFFT/cuSignal where available.
2. Data transport: Prefer 8-bit IQ for streaming/storage to reduce bandwidth unless dynamic range requirements force higher resolution captures.
3. For host interfaces, optimize host-side buffers and data paths (whether Ethernet or USB) and choose moderate sustained sample rates (e.g., 2–10 Msps per channel) for reliable real-time operation.
4. For network-attached 7020-class SDRs, offload decimation/filtering to the device SoC/FPGA and tune libiio buffer settings to prevent dropped samples.
5. UI / orchestration: Jetson host runs `iiod` client(s) and a Flask (or Django) web dashboard with gr-specest waterfalls and REST APIs.

Upgrade cost matrix and example budget

Item	**Benefit**	**Cost ($)**
7020-SDR (AD9361)	Standalone SoC architecture, GigE	200
Jetson Orin / Nano	ML / DL acceleration	269
Sync cable / GPSDO	Coherent timing and PPS/10 MHz refs	200
Antenna system (Comet + HyperLOG 7060)	Wideband reception	1,075
RF networking	Switch, LMR-400, Cat6, PoE	250
LNAs & filters	Noise floor optimization	150
Estimated total (example)		~ $1,944–3,000
RFSoC node	High-bandwidth FPGA/DAC (optional)	+5,000

Table 2: Example upgrade / procurement cost matrix

Similar projects & references: Representative examples and resources:

i) *DARPA Spectrum Collaboration Challenge (SC2)*: cognitive radio / spectrum-sharing research.

ii) *University testbeds*: Berkeley, Virginia Tech and others provide SDR testbeds and cognitive-radio research.

iii) *Commercial solutions*: ThinkRF, Per Vices (spectrum monitoring appliances).

iv) *Open-source projects:* GNU Radio, OpenAirInterface (OAI), SigMF (Signal Metadata Format), Pothos SDR, gr-specest, and various GitHub repositories with spectrum scanning and ML classifiers.

Throughput note (comparison): Practical host-interface throughput depends on the chosen hardware: typical host-limited USB 3.0 practical throughput is on the order of a few hundred MB/s; for very high-throughput needs prefer 1/10 GbE-capable platforms and appropriate network interfaces. This is a general comparison note only — host-attached USB SDRs are intentionally not itemized in the BOM/hardware lists above.

Final remarks (content parity): All material has been normalized around the network-attached 7020-class SDR and the Jetson compute node. Mentions of host-attached USB SDRs have been removed from BOM and other hardware sections; a short comparison note is included to preserve guidance for evaluation without creating vendor-specific BOM entries.